Lecture - 12

Chapter 5

IMAGE RESTORATION AND RECONSTRUCTION

Periodic Noise Reduction by Frequency Domain Filtering

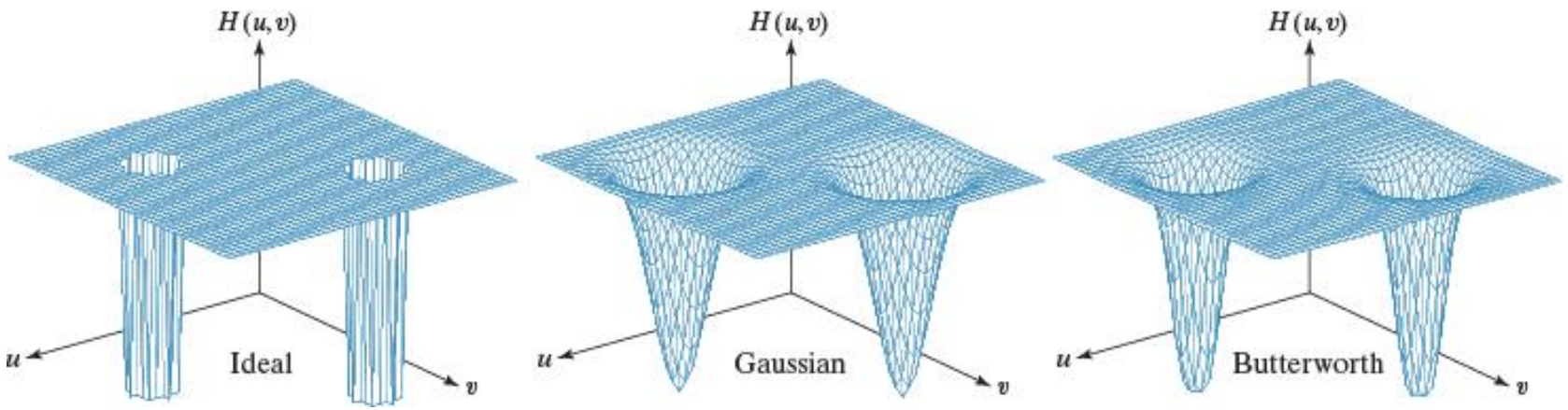
- **Bandreject, bandpass, and notch filters as tools for periodic noise reduction**

- Bandreject filters remove or attenuate a band of frequencies about the origin of the Fourier transform. Similar to those LPFs and HPFs studied, we can construct ideal, Butterworth, and Gaussian bandreject filters

- Notch filters that pass, rather than suppress: $H_{np}(u,v) = 1 - H_{nr}(u,v)$

- *NR* filters become highpass filters if $u_0 = v_0 = 0$
- *NP* filters become lowpass filters if $u_0 = v_0 = 0$

Periodic Noise Reduction by Frequency Domain Filtering



a b c

FIGURE 5.15 Perspective plots of (a) ideal, (b) Gaussian, and (c) Butterworth notch reject filter transfer functions.

Periodic Noise Reduction by Frequency Domain Filtering

Notch Pass Filtering

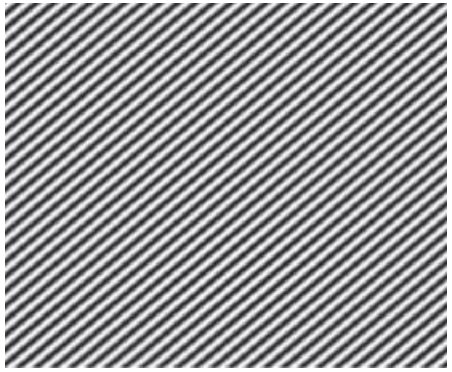
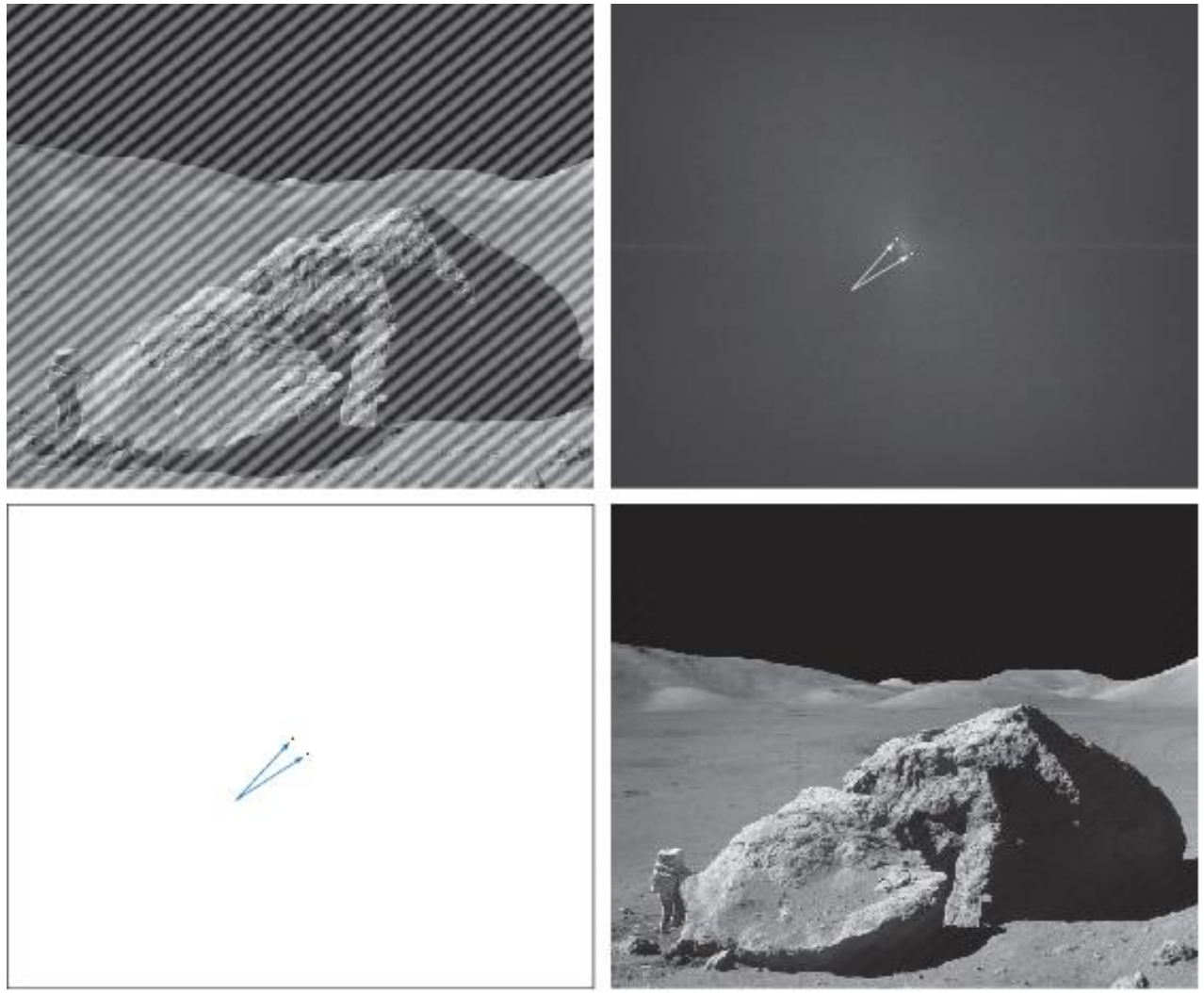


FIGURE 5.17 Sinusoidal pattern extracted from the DFT of Fig. 5.16(a) using a notch pass filter.

Notch Reject Filter



a b
c d

FIGURE 5.16 (a) Image corrupted by sinusoidal interference. (b) Spectrum showing the bursts of energy caused by the interference. (The bursts were enlarged for display purposes.) (c) Notch filter (the radius of the circles is 2 pixels) used to eliminate the energy bursts. (The thin borders are not part of the data.) (d) Result of notch reject filtering. (Original image courtesy of NASA.)

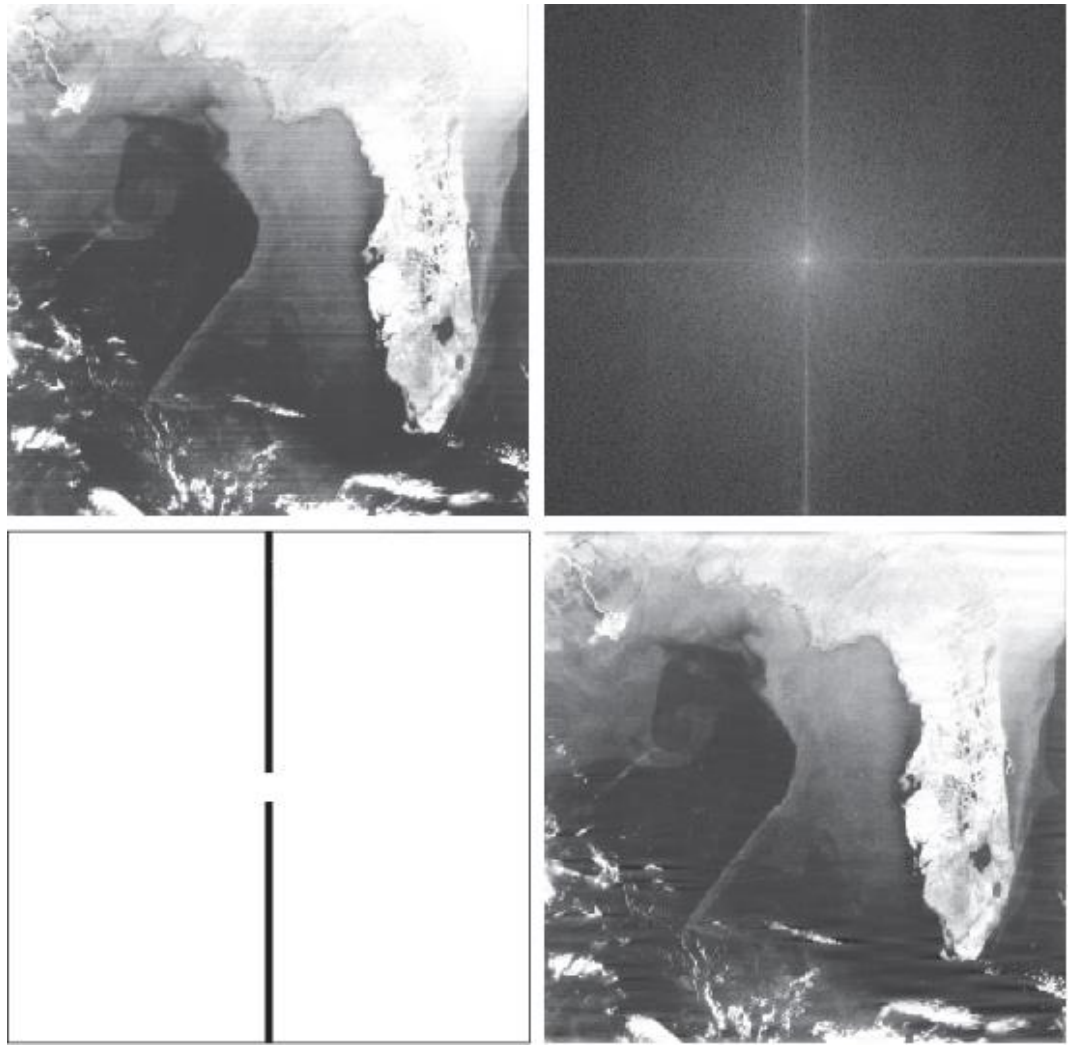
Notch Reject Filtering

Periodic Noise Reduction by Frequency Domain Filtering

Notch Pass Filtering

FIGURE 5.19 Noise pattern extracted from Fig. 5.18(a) by notch pass filtering.

Notch Reject Filter

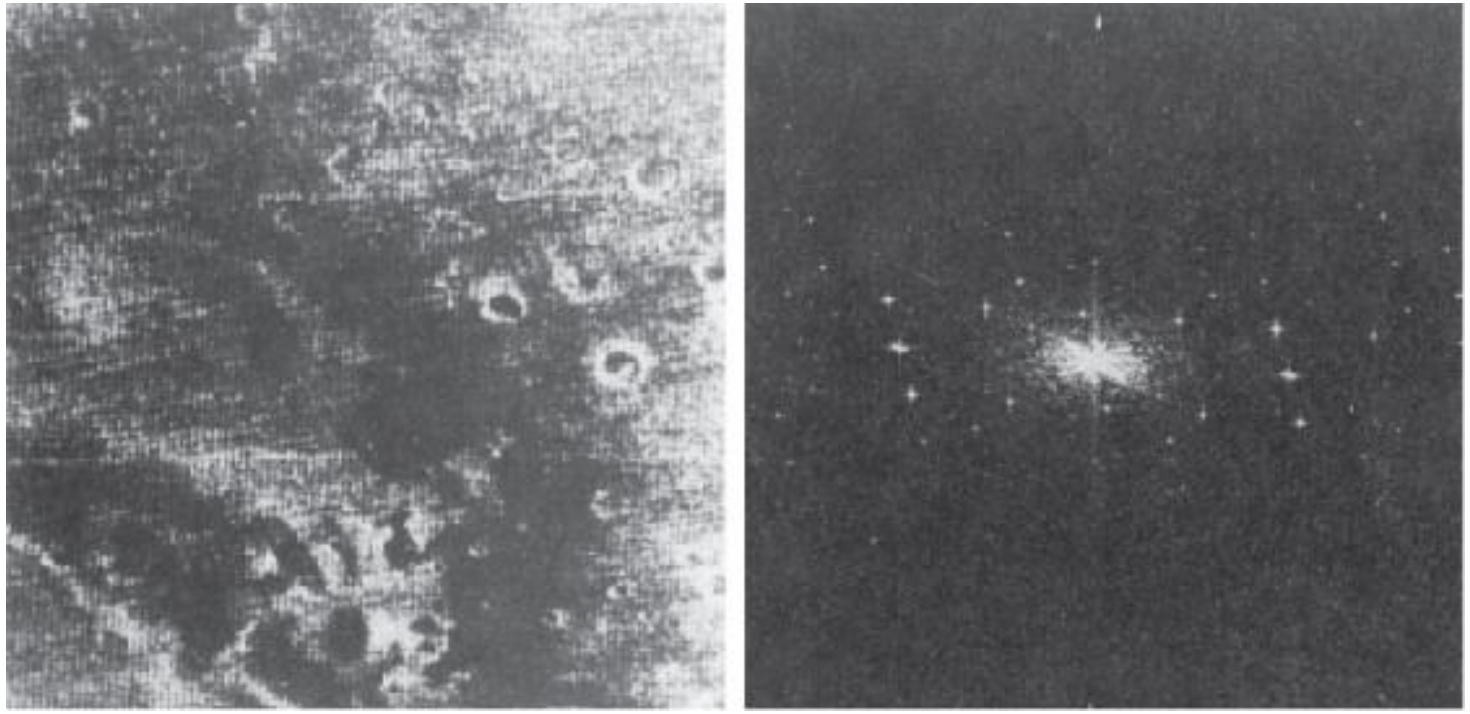


a b
c d

FIGURE 5.18 (a) Satellite image of Florida and the Gulf of Mexico. (Note horizontal sensor scan lines.) (b) Spectrum of (a). (c) Notch reject filter transfer function. (The thin black border is not part of the data.) (d) Filtered image. (Original image courtesy of NOAA.)

Notch Reject Filtering

Optimum Notch Filtering



a b

FIGURE 5.20 (a) Image of the Martian terrain taken by Mariner 6. (b) Fourier spectrum showing periodic interference. (Courtesy of NASA.)

Optimum Notch Filtering

- In the ideal case, the original image can be restored if the noise can be estimated completely.
  - That is:

$$f(x,y) = g(x,y) - \eta(x,y)$$

- However, the noise can be only partially estimated. This means the restored image is not exact.
  - Which means

$$\hat{f}(x,y) = g(x,y) - \hat{\eta}(x,y)$$

Optimum Notch Filtering

- In this section, we try to improve the restored image by introducing a modulation function

$$\hat{f}(x,y) = g(x,y) - w(x,y)\hat{\eta}(x,y)$$

  - Here the modulation function is a constant within a neighborhood of size (2a+1) by (2b+1) about a point (x,y)
  - We optimize its performance by minimizing the local variance of the restored image at the position (x,y)

$$\sigma^2(x,y) = \frac{1}{(2a+1)(2b+1)} \sum_{s=-a}^{a} \sum_{t=-b}^{b} \left[\hat{f}(x+s, y+t) - \bar{\hat{f}}(x,y) \right]^2$$

$$\bar{\hat{f}}(x,y) = \frac{1}{(2a+1)(2b+1)} \sum_{s=-a}^{a} \sum_{t=-b}^{b} \hat{f}(x+s, y+t)$$

Optimum Notch Filtering

$$\sigma^2(x,y) = \frac{1}{(2a+1)(2b+1)} \sum_{s=-a}^{a} \sum_{t=-b}^{b} \{[g(x+s,y+t) - w(x+s,y+t)\hat{\eta}(x+s,y+t)] - [\bar{g}(x,y) - \overline{w(x,y)\hat{\eta}(x,y)}]\}^2$$

Assumption: $w(x+s,y+t) = w(x,y) \quad for \ -a \le s \le a \ and \ -b \le t \le b$

$$\Rightarrow \ \overline{w(x,y)\hat{\eta}(x,y)} = w(x,y)\overline{\hat{\eta}}(x,y)$$

Optimum Notch Filtering

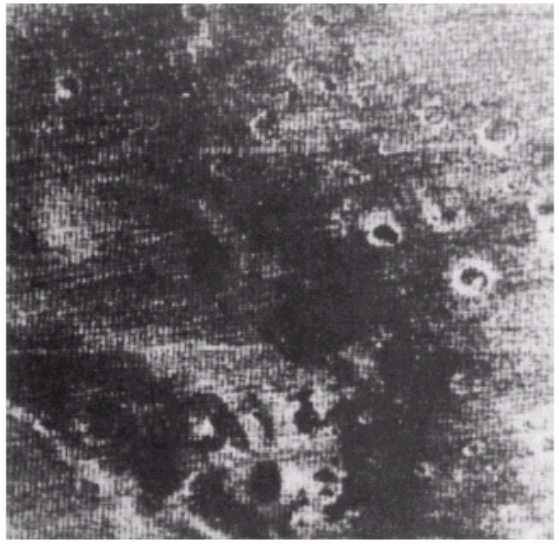
$$\sigma^2(x,y) = \frac{1}{(2a+1)(2b+1)} \sum_{s=-a}^{a} \sum_{t=-b}^{b} \{[g(x+s,y+t) - w(x+s,y+t)\hat{\eta}(x+s,y+t)] - [\bar{g}(x,y) - w(x,y)\bar{\hat{\eta}}(x,y)\}^2$$

To minimize $\sigma^2(x,y)$

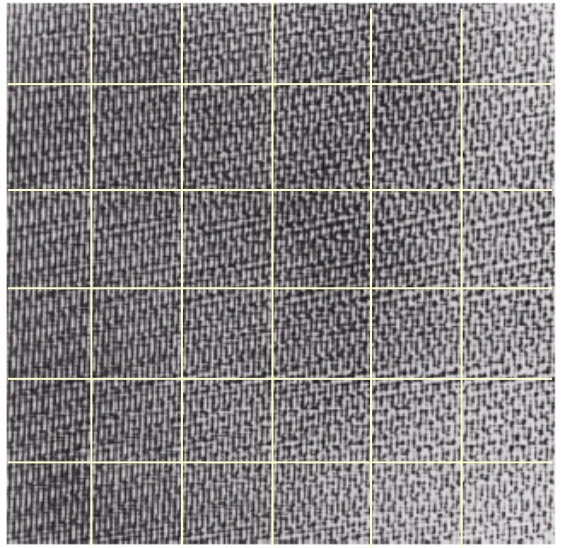
$$\frac{\partial \sigma^2(x,y)}{\partial w(x,y)} = 0$$

$$\Rightarrow w(x,y) = \frac{\overline{g(x,y)\hat{\eta}(x,y)} - \bar{g}(x,y)\bar{\hat{\eta}}(x,y)}{\overline{\hat{\eta}^2}(x,y) - \bar{\hat{\eta}}^2(x,y)}$$

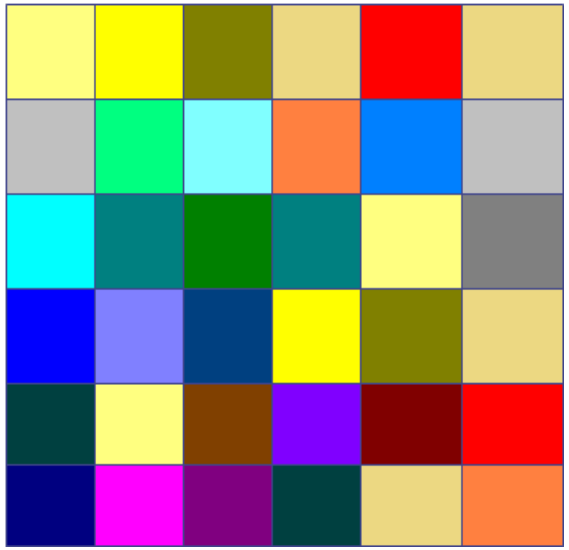
Optimum Notch Filtering



$g(x, y)$



$\hat{\eta}(x, y)$



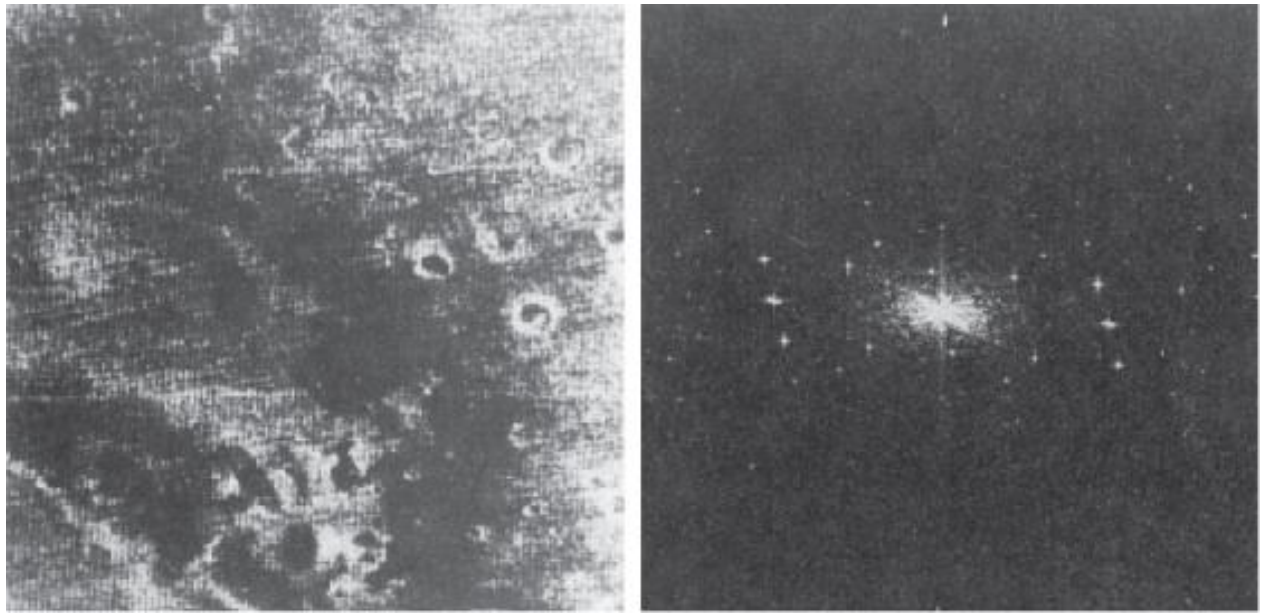
$w(x, y)$

$$\hat{f}(x, y) = g(x, y) - w(x, y)\hat{\eta}(x, y)$$

Optimum Notch Filtering

a b

FIGURE 5.20 (a) Image of the Martian terrain taken by Mariner 6. (b) Fourier spectrum showing periodic interference. (Courtesy of NASA.)



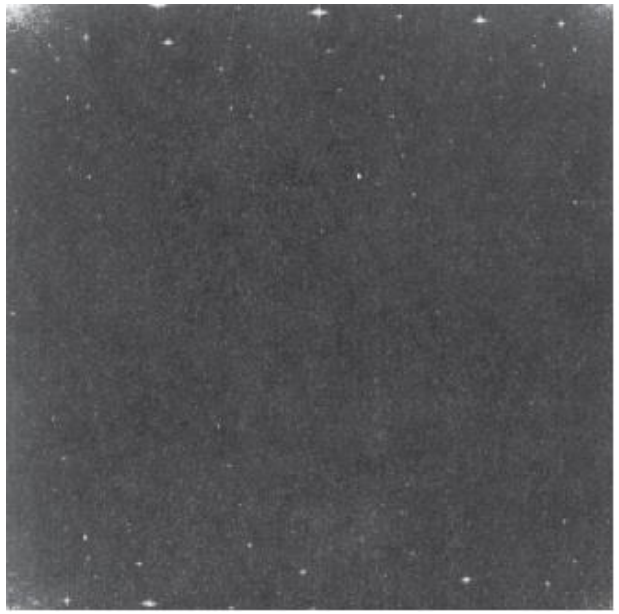


FIGURE 5.21 Uncentered Fourier spectrum of the image in Fig. 5.20(a). (Courtesy of NASA.)

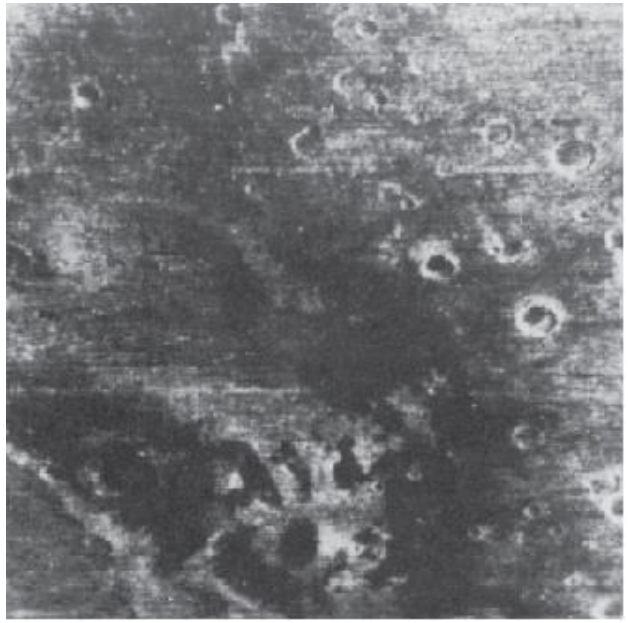
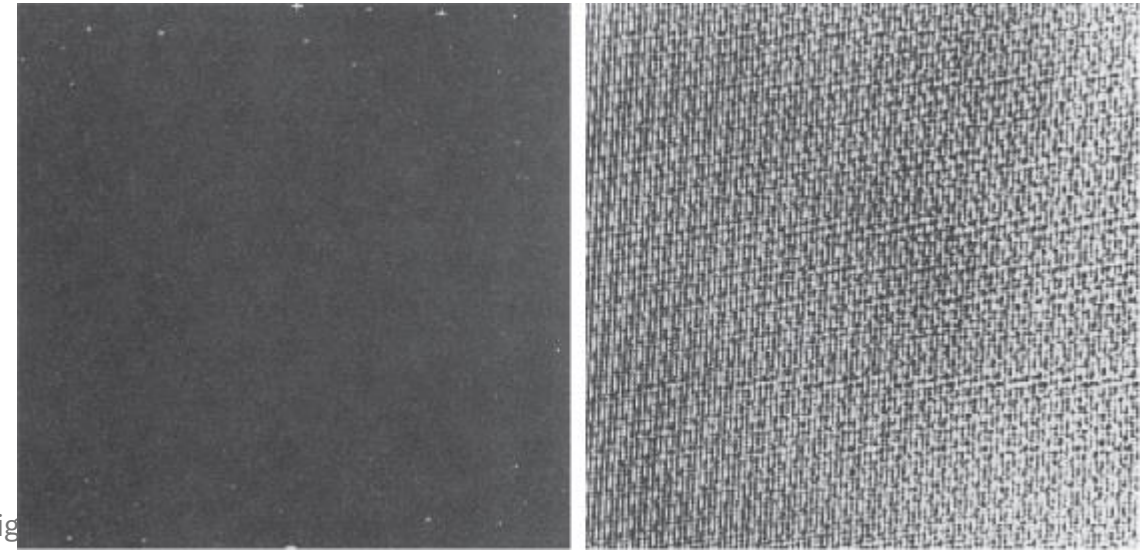


FIGURE 5.23 Restored image. (Courtesy of NASA.)



a b

FIGURE 5.22 (a) Fourier spectrum of $N(u,v)$, and (b) corresponding spatial noise interference pattern, $\eta(x,y)$. (Courtesy of NASA.)

Estimating the **Image Degradation** Function

- Degradation Model (with noise)

$$g(x, y) = H[f(x, y)] + \eta(x, y)$$

- **Principal way to estimate the degradation function for use in image restoration:**
  - **Observation**
  - **Experimentation**
  - **Mathematical modeling**

Estimating by Image Observation

- We look for a small section of the image that has strong signal content ($g_s(x,y)$) and then construct an un-degradation of this section by using sample gray levels ($\hat{f}_s(x,y)$).

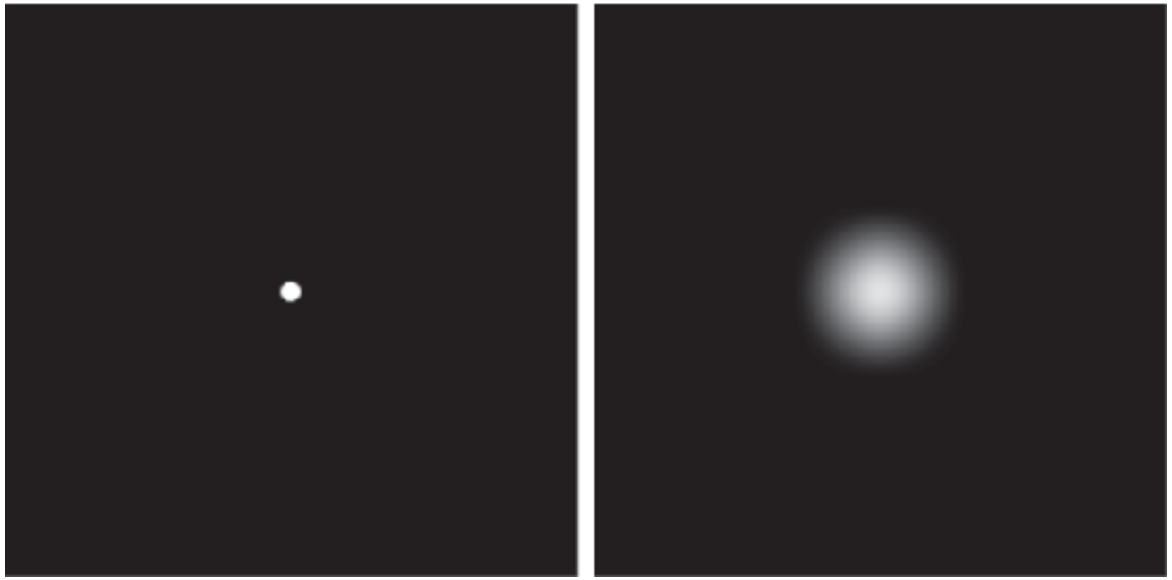
$$H_s(u,v) = \frac{G_s(u,v)}{\hat{F}_s(u,v)}$$

- Now, we construct a function $H(u,v)$ on a large scale, but having the same shape.

Estimating by Experimentation

- We try to obtain impulse response of the degradation by imaging an impulse (small dot of light) using the system. Therefore

$$H(u,v) = \frac{G(u,v)}{A}$$



a b

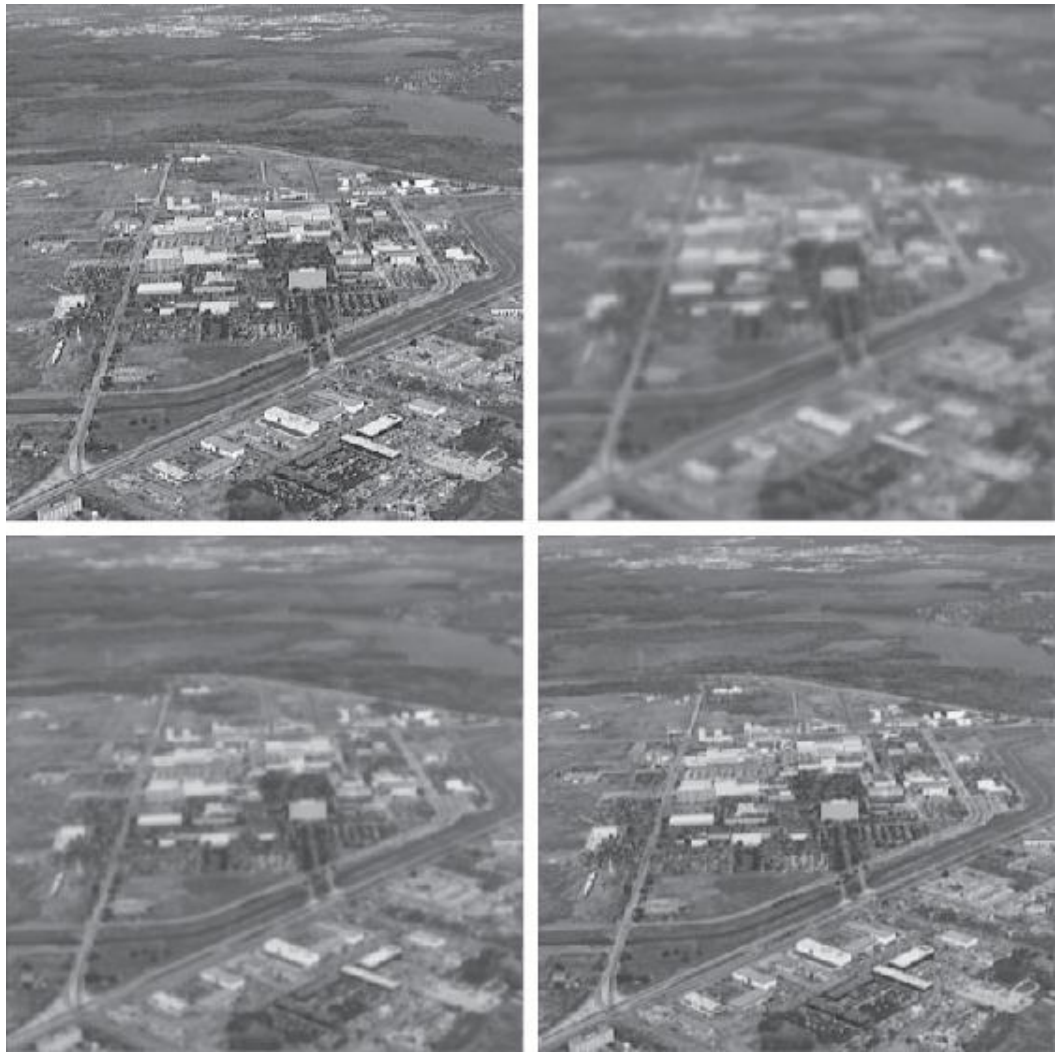
FIGURE 5.24 Estimating a degradation by impulse characterization. (a) An impulse of light (shown magnified). (b) Imaged (degraded) impulse.

Estimating by Modeling

- Atmospheric turbulence model: $H(u,v) = e^{-k(u^2+v^2)^{5/6}}$

Severe turbulence k=0.0025

Negligible turbulence



a b
c d

FIGURE 5.25 Modeling turbulence. (a) No visible turbulence. (b) Severe turbulence, $k = 0.0025$. (c) Mild turbulence, $k = 0.001$. (d) Low turbulence, $k = 0.00025$. All images are of size 480×480 pixels. (Original image courtesy of NASA.)

Mild turbulence k=0.001

Low turbulence k=0.00025

Estimating by Modeling

Blurring by linear motion:

$$g(x,y) = \int_0^T f[x - x_0(t), y - y_0(t)]dt$$

$$G(u,v) = F(u,v)\int_0^T e^{-j2\pi[ux_0(t)+vy_0(t)]}dt$$

$$\Rightarrow H(u,v) = \int_0^T e^{-j2\pi[ux_0(t)+vy_0(t)]}dt$$

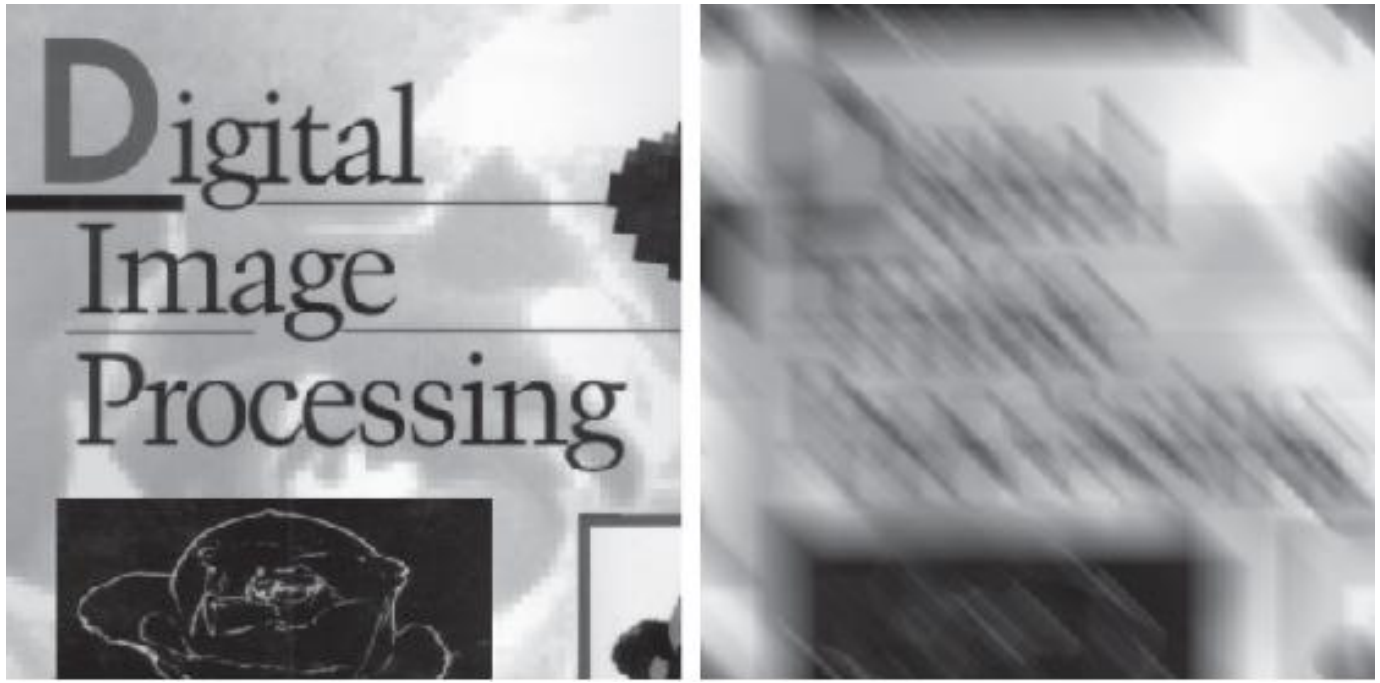
$$if \;\; x_0(t) = at/T \;\; and \;\; y_0(t) = 0 \;\Rightarrow\; H(u,v) = \int_0^T e^{-2\pi uat/T}dt$$

$$= \frac{T}{\pi ua}\sin(\pi ua)e^{-j\pi ua}$$

Estimating by Modeling

$$if \;\; x_0(t) = at/T \;\; and \;\; y_0(t) = bt/T \;\; \Rightarrow$$

$$H(u,v) = \frac{T}{\pi(ua+vb)} \sin[\pi(ua+vb)] e^{-j\pi(ua+vb)}$$



a b

FIGURE 5.26 (a) Original image. (b) Result of blurring using the function in Eq. (5-77) with $a = b = 0.1$ and $T = 1$.

Inverse Filtering

- The simplest approach to restoration is direct inverse filtering:

$$\hat{F}(u,v) = \frac{G(u,v)}{H(u,v)}$$

$$\hat{F}(u,v) = F(u,v) + \frac{N(u,v)}{H(u,v)}$$

Even if we know the degradation function, we cannot recover the un-degraded image

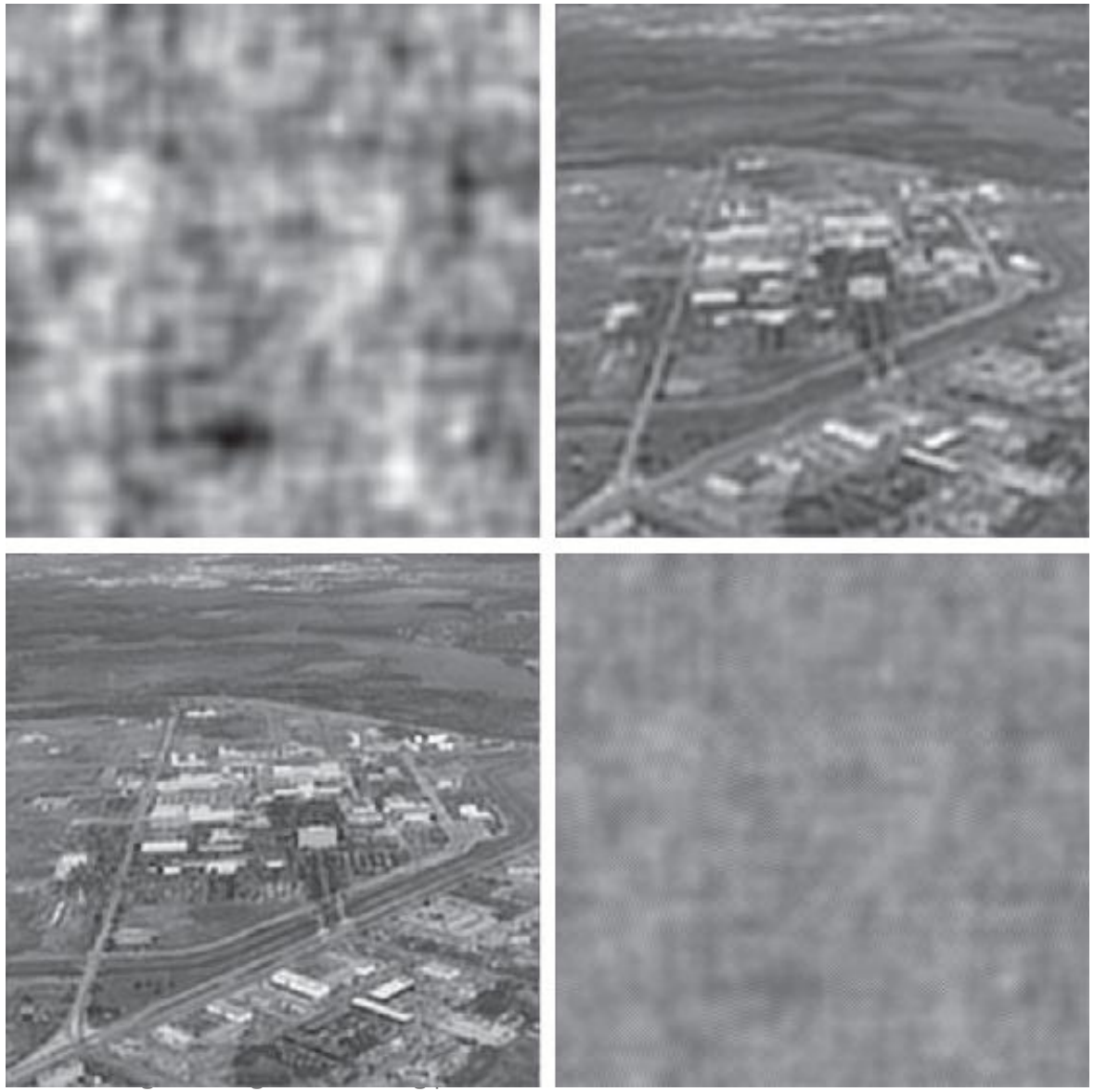
If the degradation has zero or very small values, then the ratio *N/H could easily dominate our estimation of F*.

One approach to get around the zero or small-value problem is to limit the filter frequencies to value near the origin.

Inverse Filtering

Degraded Image

40



70

85

a b
c d

FIGURE 5.27 Restoring Fig. 5.25(b) using Eq. (5-78). (a) Result of using the full filter. (b) Result with H cut off outside a radius of 40. (c) Result with H cut off outside a radius of 70. (d) Result with H cut off outside a radius of 85.

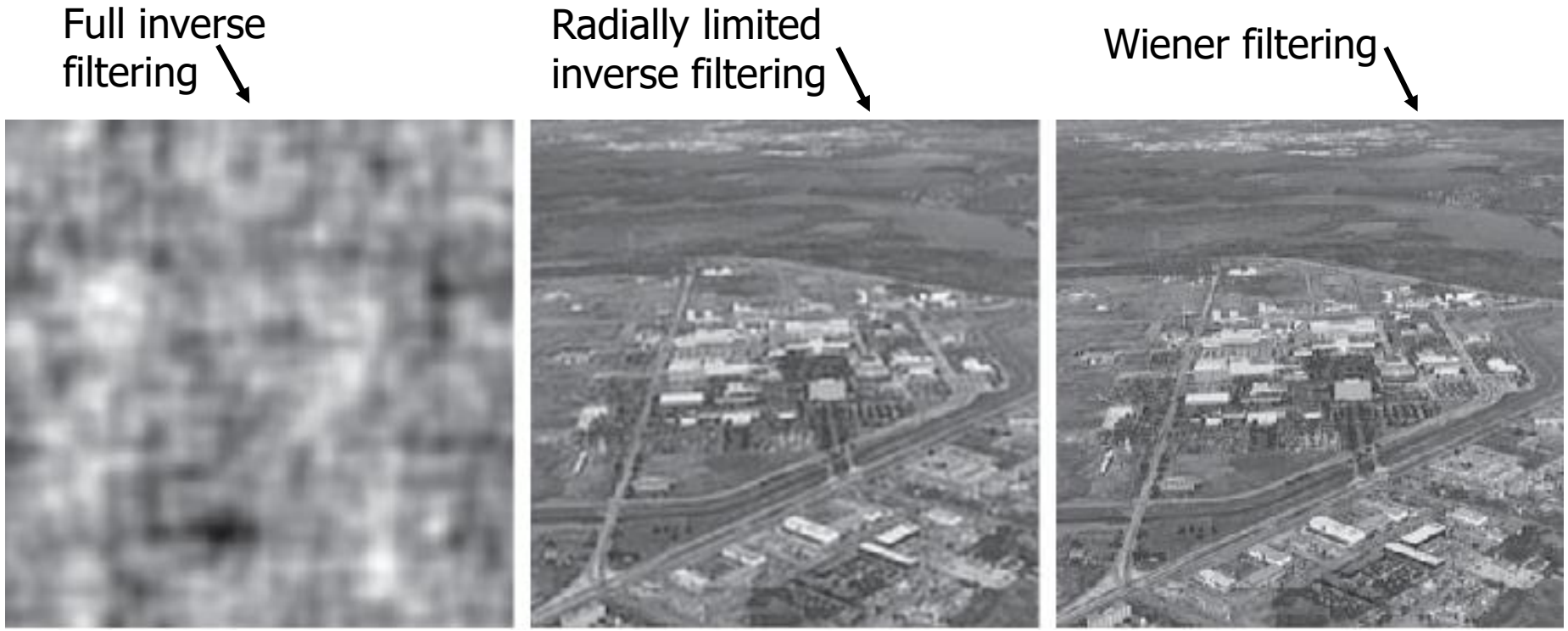
Minimum Mean Square Error Filtering (Wiener Filtering)

This approach incorporate both the degradation function and statistical characteristic of noise into the restoration process.

$$e^2 = E[(f - \hat{f})^2]$$

The objective is to find an estimation for *f* such that minimized e^2

Wiener Filtering



a b c

FIGURE 5.28 Comparison of inverse and Wiener filtering. (a) Result of full inverse filtering of Fig. 5.25(b). (b) Radially limited inverse filter result. (c) Wiener filter result.

Wiener Filtering

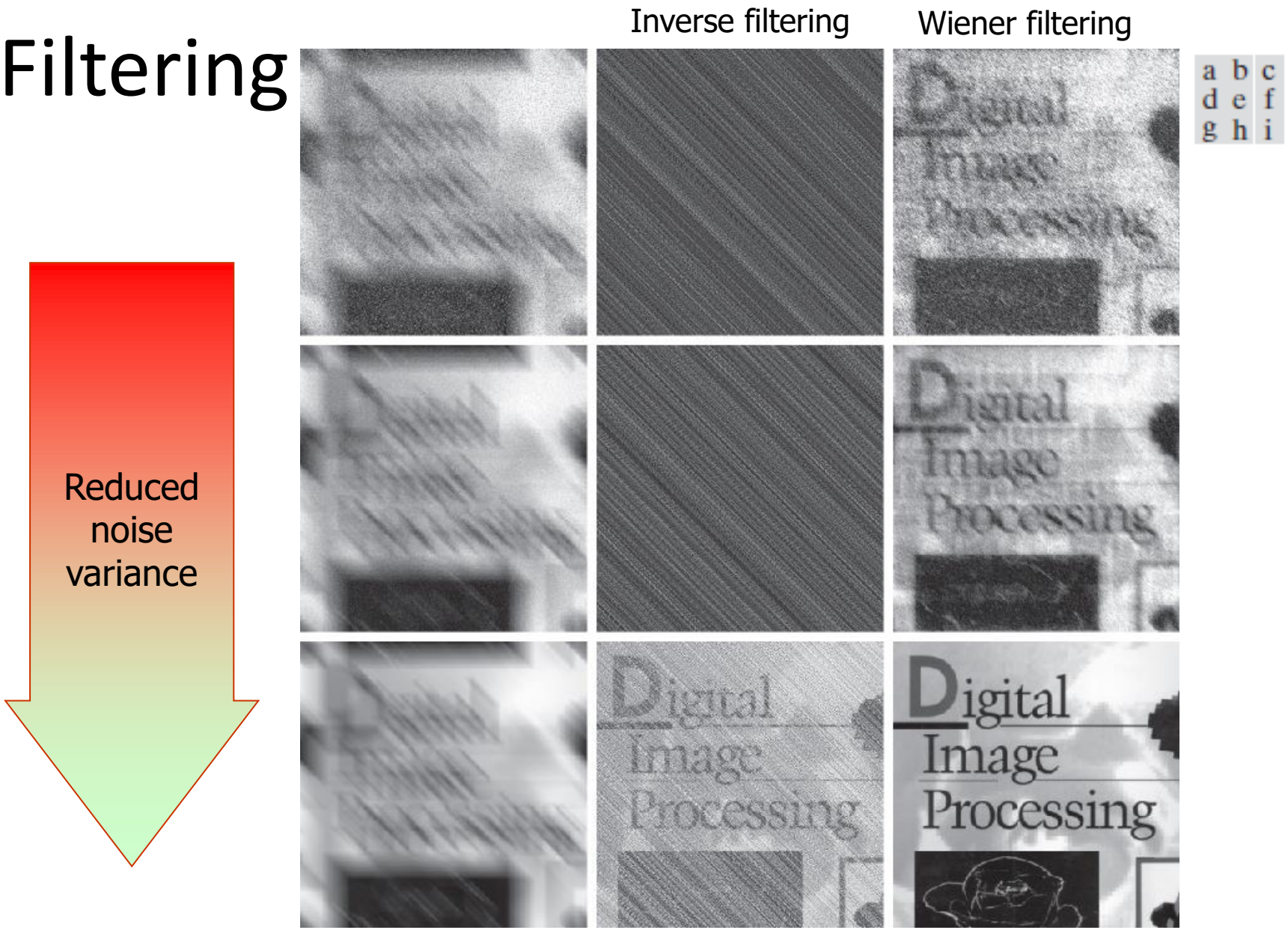


FIGURE 5.29 (a) 8-bit image corrupted by motion blur and additive noise. (b) Result of inverse filtering. (c) Result of Wiener filtering. (d)–(f) Same sequence, but with noise variance one order of magnitude less. (g)–(i) Same sequence, but noise variance reduced by five orders of magnitude from (a). Note in (h) how the deblurred image is quite visible through a "curtain" of noise.